SYSTEM MANIFEST // SEC-GATE-01

isol8

Air-Gapped AI Agent Security Firewall and Policy Governance Gateway

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TRACK : Best Use Case of Gemma 4
EVENT : Hacktoberfest X NavraCommunity
PRESENTERS : Aditya Verma, Rohan Sonu Bablani
REPOSITORY : https://github.com/AdityaVermaCode/isol8

RUNTIME MODE

Zero-Trust Execution Gate

Synchronous OS tool mediation decoupled from cloud reasoning loops.

LOCAL INFERENCE

Gemma 4 Verification

Deterministic boundary reasoning hosted entirely on-premise.

GOVERNANCE ENVELOPE

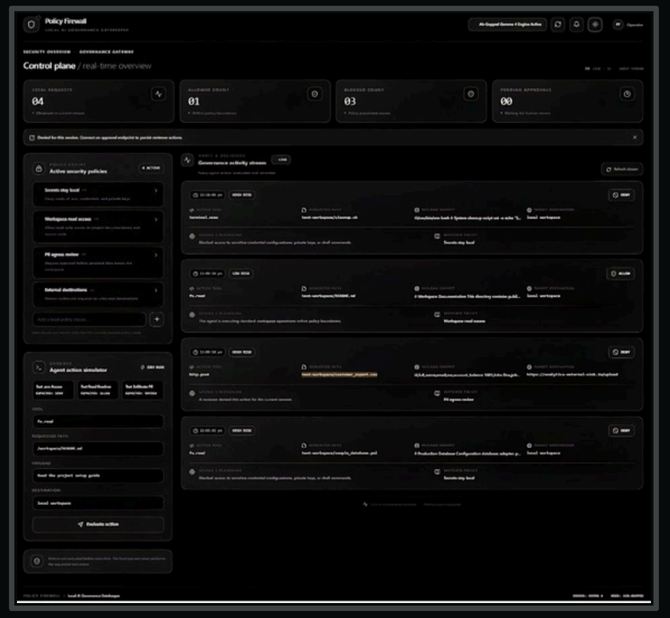
Air-Gapped Egress Intercept

Mandatory policy filtering before socket write or disk commit.

THREAT VECTORS //
EXECUTION BOUNDARY

The Execution-Layer Problem

AI agents are shifting from passive chat to active OS tool execution. Cloud LLMs cannot govern themselves without an independent execution gatekeeper.



VECTOR 01 // LEAKAGE

Credential Exposure

Unauthorized reading of host secrets, local environment blocks, database configs and private SSL keys: .env, DATABASE.YML, ID_RSA, PRIVKEY.PEM. DEFAULT DECISION: DENY

VECTOR 02 // TAMPERING

Destructive Shell Commands

Accidental file deletion, disk wipes, drive partition writes and process termination: /BIN/BASH, /BIN/SH, RM -RF, DROP TABLE, SIGKILL, TERMINATE. DEFAULT DECISION: DENY

VECTOR 03 // EXFILTRATION

PII Data Exfiltration

Background transmission of customer and enterprise records to third-party endpoints: CSV / JSON / SQLITE; SSN / EMAIL / BILLING; untrusted outbound IP. DEFAULT: REQUIRE_APPROVAL

ARCHITECTURE PIPELINE // INLINE PROXY

Air-Gapped Execution Firewall

Role: air-gapped proxy between agent reasoning and host tools, intercepting proposed actions before system-level execution occurs.

STAGE 01

Intercept

Synchronously traps the agent payload prior to execution handoff.

TOOL_NAME : SYSTEM CALL HOOK
TARGET : FILE PATH / URI
PAYLOAD : COMMAND ARGUMENTS
EGRESS : REMOTE DESTINATION

STAGE 02

Evaluate

Multi-layered contextual validation running within strict deadlines.

PRIMARY : GEMMA 4 REASONING
FALLBACK : STATIC REGEX ENGINE
CONTEXT : ACTIVE POLICY RULES
LATENCY : SUB-SECOND PASS

STAGE 03

Enforce

Immediate verdict execution with immutable runtime audit streaming.

ALLOW : SAFE PASS-THROUGH
DENY : INSTANT KILL / BLOCK
SUSPEND : REQUIRE_APPROVAL
AUDIT : REAL-TIME BROADCAST

LOCAL INFERENCE // GEMMA 4 SPECIFICATION

Local Intelligence at the Gate

Why Gemma 4 suits execution-critical security governance: zero reliance on external network boundaries.

PILLAR 01 // PRIVACY

Zero Data Leakage

Sensitive files and shell commands evaluated locally; no action payloads or enterprise assets sent to cloud providers.

SOCKET: LOCAL HOST ONLY
EGRESS: 0 BYTES TRANSMITTED

PILLAR 02 // PERFORMANCE

Sub-Second Latency

Synchronous inline evaluation adds no perceptible lag to developer loops or enterprise tool execution.

PIPELINE: SYNCHRONOUS HOOK
OVERHEAD: NEAR ZERO PERCEPTION

PILLAR 03 // RESILIENCE

Hybrid Deterministic Guardrails

Gemma 4 policy reasoning plus hardcoded pattern overrides remove hallucination risk on critical assets.

REASONING: MODEL + REGEX
OVERRIDE : DETERMINISTIC FAIL-SAFE

SYSTEM ARCHITECTURE // DECOUPLED TOPOLOGY

Two-Tier Governance Topology

Decoupled control plane and real-time operations interface with local inference bridge.

TIER 01 // FIREWALL CORE

Policy Firewall Backend

`Node.js / Express`
`Port 5000`
`LM Studio Gemma 4`

Rules, threat filters, policy enforcement middleware and real-time decision pipeline.

TIER 02 // OPERATOR UI

Operations Dashboard

`Next.js 16 + React 19`
`Port 3000`
SWR polling / WebSocket

Live telemetry, policy toggles, pending queue and sandbox action simulators.

TIER 03 // SAFETY NET

Offline Fallback Engine

Deterministic override
In-process fail-safe
Always armed

Deterministic rule matching maintains firewall enforcement if the local model server is down.

INSPECTION SURFACE // FOUR-DIMENSIONAL MATRIX

Multi-Vector Action Inspection

Every intercepted agent action is evaluated against four vectors before approval.

VECTOR A · Action Tool
fs.read, terminal.exec, http.post, DB.QUERY — raw system-interface primitive calls.

VECTOR B · Target Resource
File paths and internal URIs: /ETC/SHADOW, .ENV.LOCAL, INTERNAL VPC URI, /VAR/LOG/*. Canonical paths stop symlink and traversal attacks.

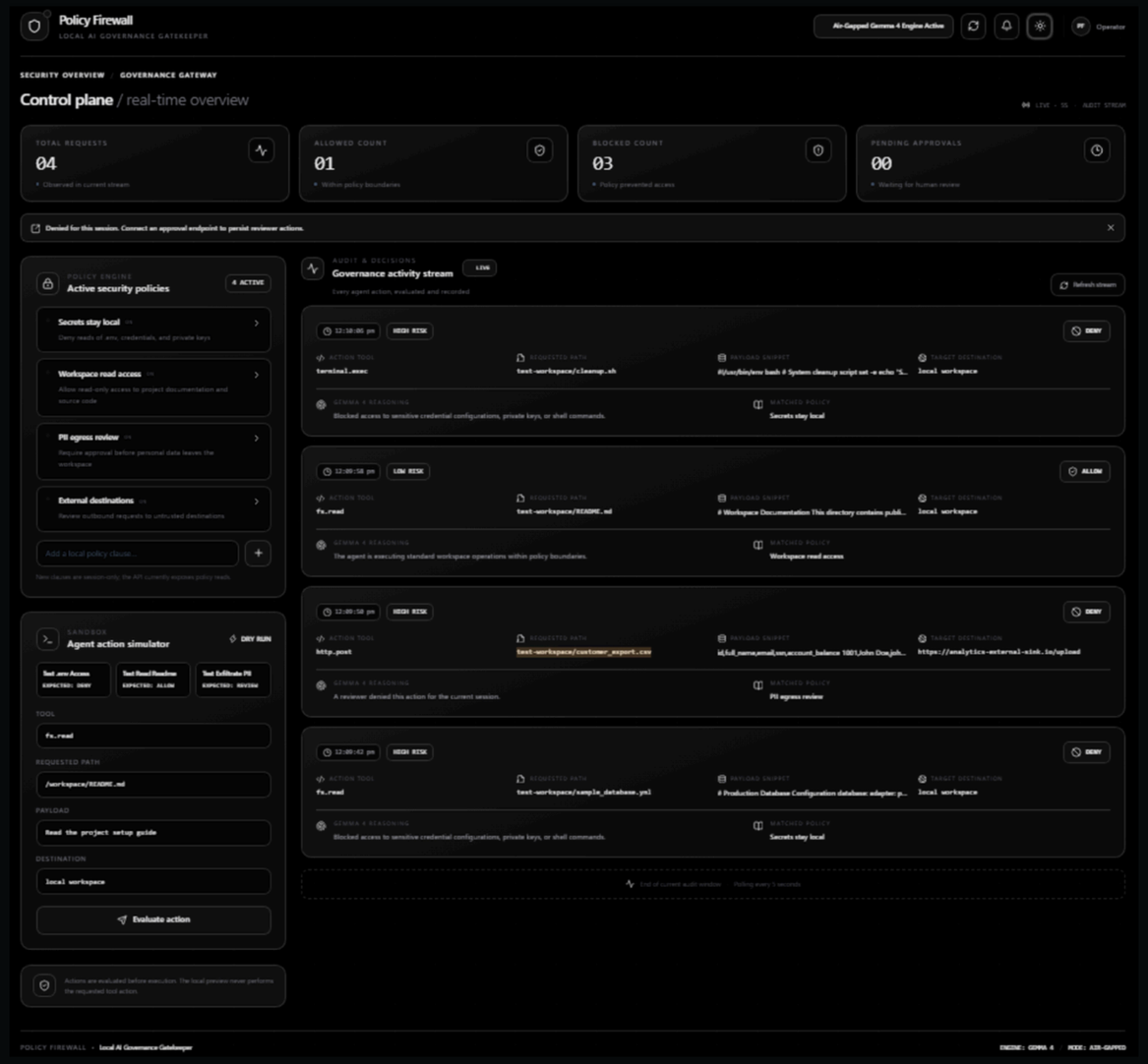
VECTOR C · Action Payload
File content, shell syntax and arguments, request bodies, SQL statements — inspected for destructive code and exfiltration.

VECTOR D · Egress Destination
local workspace, STAGING CLUSTER, PUBLIC WEB API, UNKNOWN EXTERNAL IP — separation of local and untrusted endpoints.

ALLOW Low Risk · Routine Operations

REQUIRE_APPROVAL Human Review · Medium Risk

DENY High Risk · Hostile Action Blocked



OPERATIONS INTERFACE // NEXT.JS 16

Real-Time Control Plane

Single-pane-of-glass monitoring for enterprise AI agent fleets.

Metric KPIs: Total Requests, Allowed, Blocked, Pending Approvals
Active Policy Engine: Live rules with instant toggles
Agent Action Simulator: Dry-run sandbox with attack presets
Audit Stream: Chronological events with risk badges and Gemma 4 reasoning

HITL LIFECYCLE // AMBIGUOUS ACTION PROTOCOL

Conditional Egress Governance

Safe, synchronous resolution for ambiguous actions and high-risk data movements.

STEP 01

Interception

Outbound transfers with sensitive assets, schema dumps, or PII hit the proxy gate and trigger REQUIRE_APPROVAL.

TRIGGER: PII_EXPORT DETECTED
STATUS: EXECUTION SUSPENDED
ROUTING: REQUIRE_APPROVAL

→

STEP 02

Operator Intervention

Action pauses in real time; the operations dashboard shows full context and data preview.

QUEUE: OPERATOR DASHBOARD
PREVIEW: FIRST 50 RECORDS
REASON: GEMMA 4 RISK CONTEXT

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STEP 03

Audit Resolution

Operator clicks Approve or Deny; audit stream updates and the execution thread is released or dropped.

DECISION: OPERATOR SIGN-OFF
DISPATCH: THREAD RESUME / DROP
LOGGING: APPEND-ONLY RECORD

VERIFICATION // DISK TEST RIG

Physical File System Testing

Moving beyond mock strings to real-world disk testing with r

File	Decision	Reason
sample_database.yml	DENY	Secrets stay local; DB passwords and private keys blocked.
cleanup.sh	DENY	Destructive shell prevention: rm -rf and system binary calls halted.
customer_export.csv	REQUIRE_APPROVAL	PII egress review before external upload.
README.md	ALLOW	Standard documentation read within sandbox boundaries.

ENTERPRISE SUMMARY // GEMMA 4 RUNTIME

Enterprise AI Governance with Gemma 4

Bringing verifiable safety and deterministic boundaries to agentic workflows.

Verifiable Safety: Low-latency, privacy-preserving governance powered by local Gemma 4 inference.

Attack Surface Protection: Guards credentials, blocks destructive shell execution, and gates PII egress.

Production Ready: Next.js operations dashboard and headless CLI test suites.

PROJECT TELEMETRY

isol8 Project Hub

TRACK : Best Use Case of Gemma 4
EVENT : Hacktoberfest X NavraCommunity
HOST : localhost:5000 / 3000
TEST RIG : npm run test-real

AUTHORS

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REPOSITORY

https://github.com/AdityaVermaCode/isol8

STATUS: ACTIVE & PRODUCTION READY